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(54) **PULSE WIDTH CONTROL APPARATUS AND METHOD**

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(57)

ABSTRACT

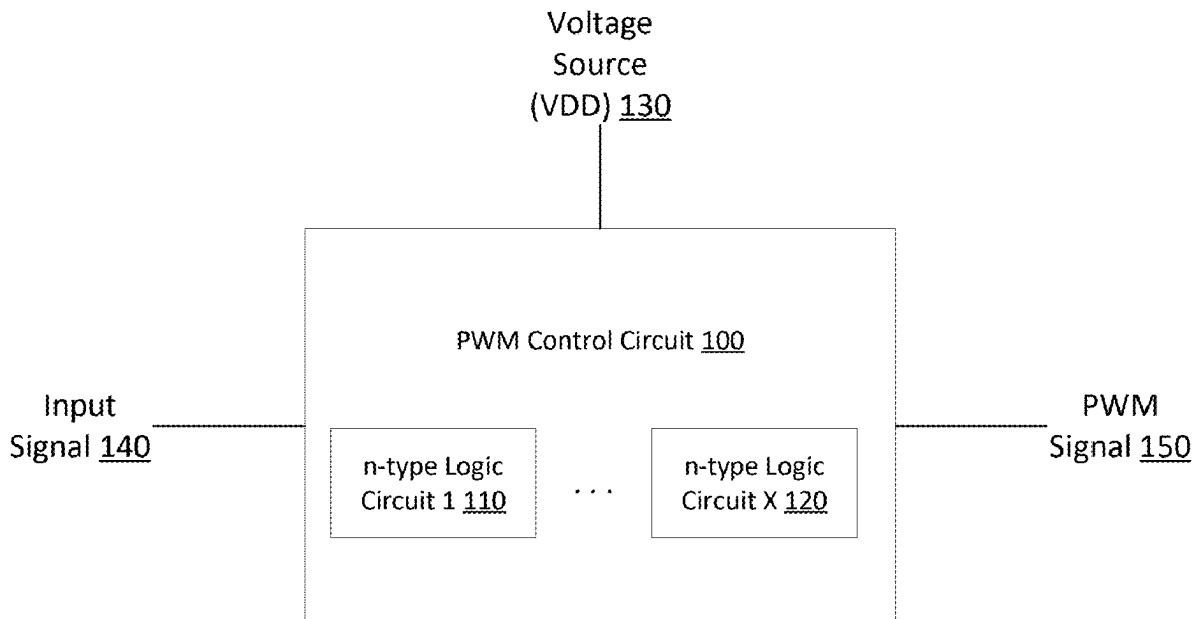
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(63) Continuation of application No. 18/366,991, filed on Aug. 8, 2023, now Pat. No. 12,334,925, which is a continuation of application No. 17/836,046, filed on Jun. 9, 2022, now Pat. No. 11,764,760.

(60) Provisional application No. 63/329,942, filed on Apr. 12, 2022.

Systems, methods, and devices are described herein for generating a pulse width modulation (PWM) signal having a specific duty cycle. In one embodiment, the system includes a square wave generator and a logic device. The square wave generator is configured to delay a input square wave signal to generate a plurality of square wave signals. The logic device is configured to perform a logic operation to two of square wave signals of the plurality of square wave signals, which in turn generates the PWM signal having a duty cycle corresponding to the two square wave signals.



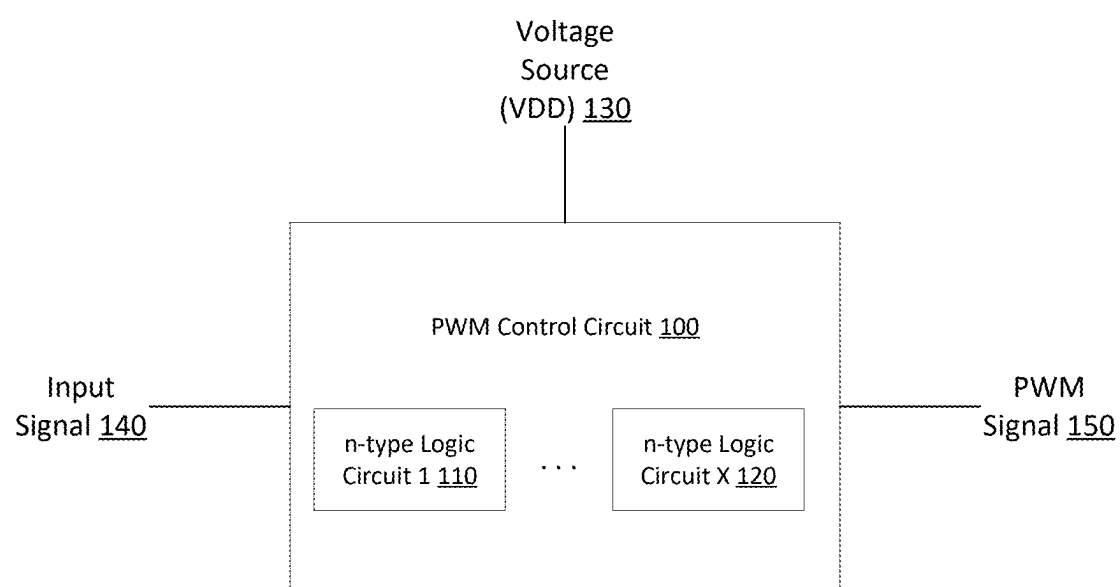


FIG. 1

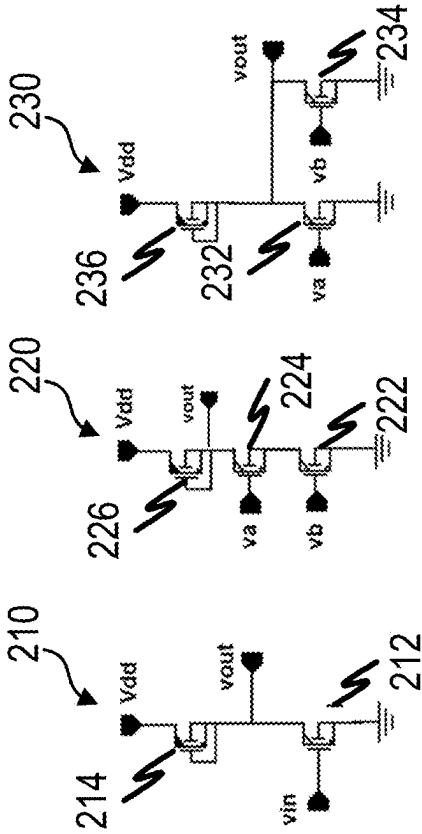


FIG. 2A

FIG. 2B

FIG. 2C

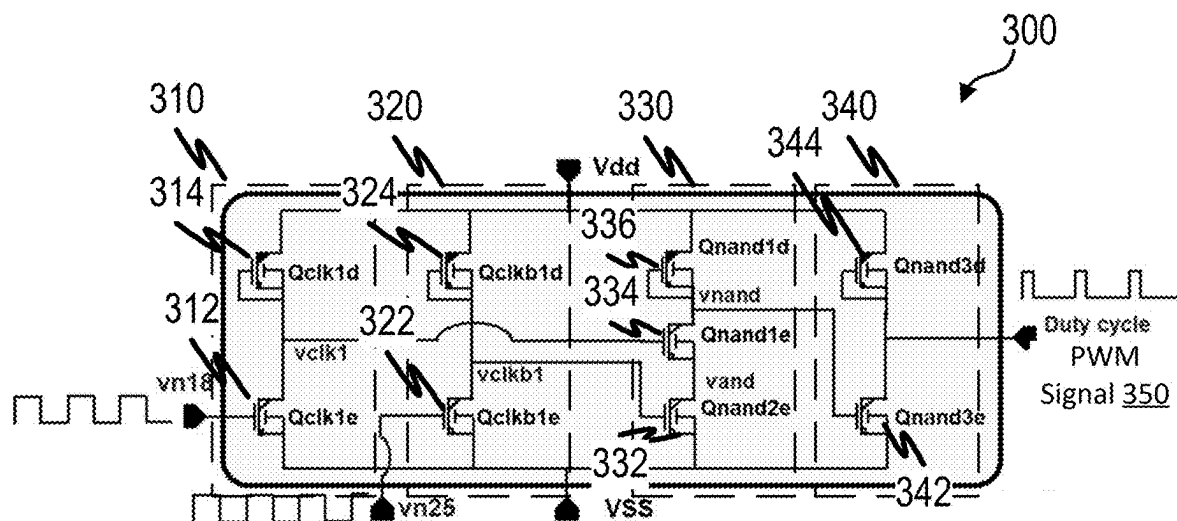


FIG. 3A

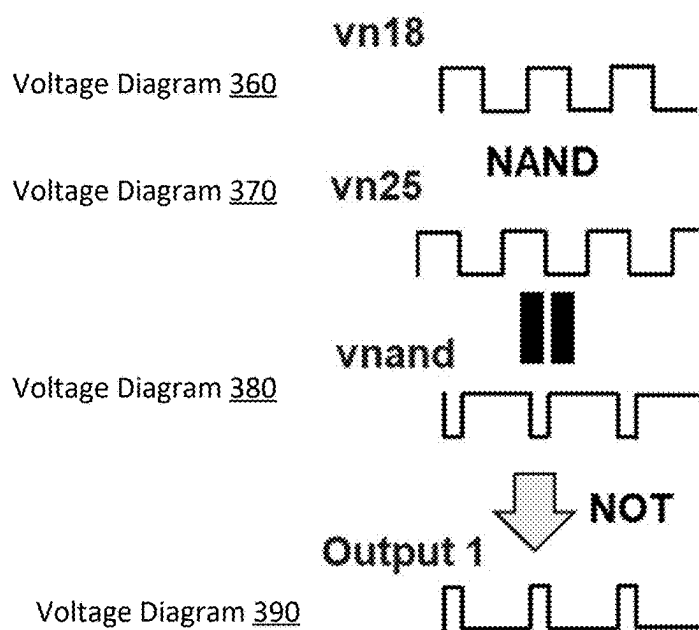


FIG. 3B

212

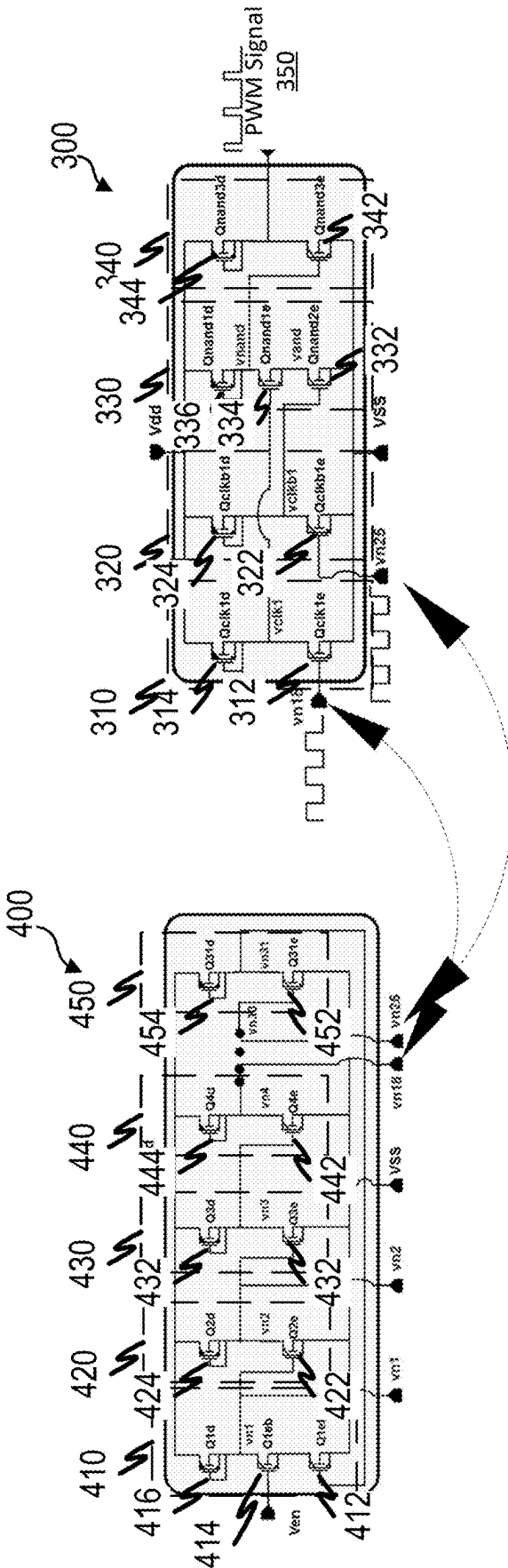


FIG. 4

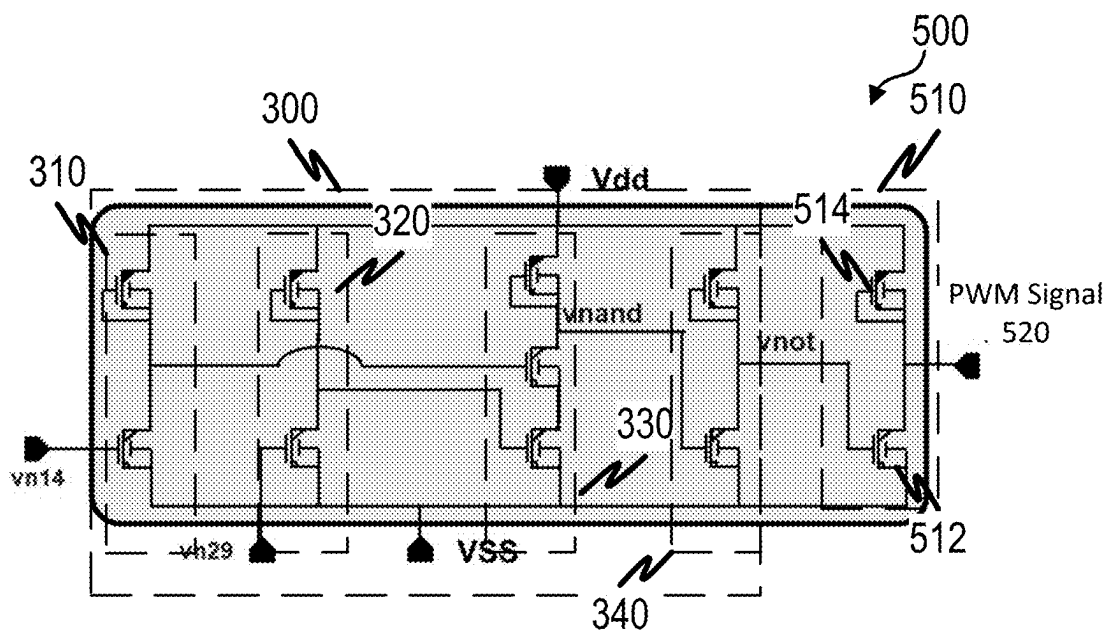


FIG. 5A

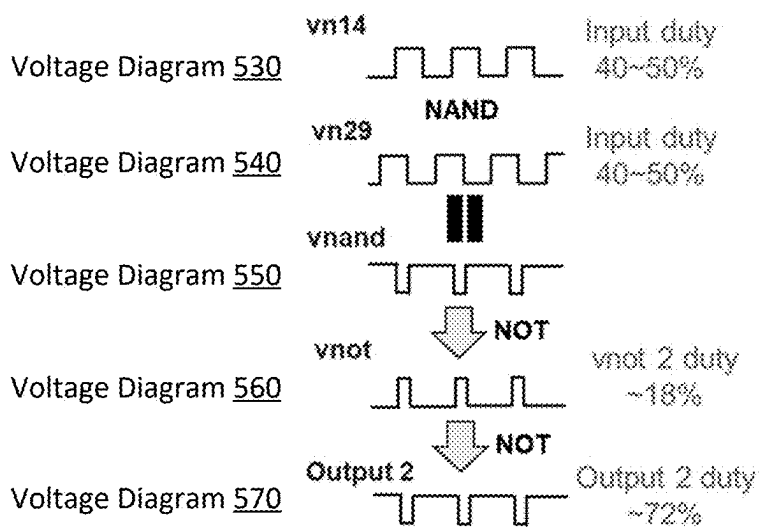


FIG. 5B

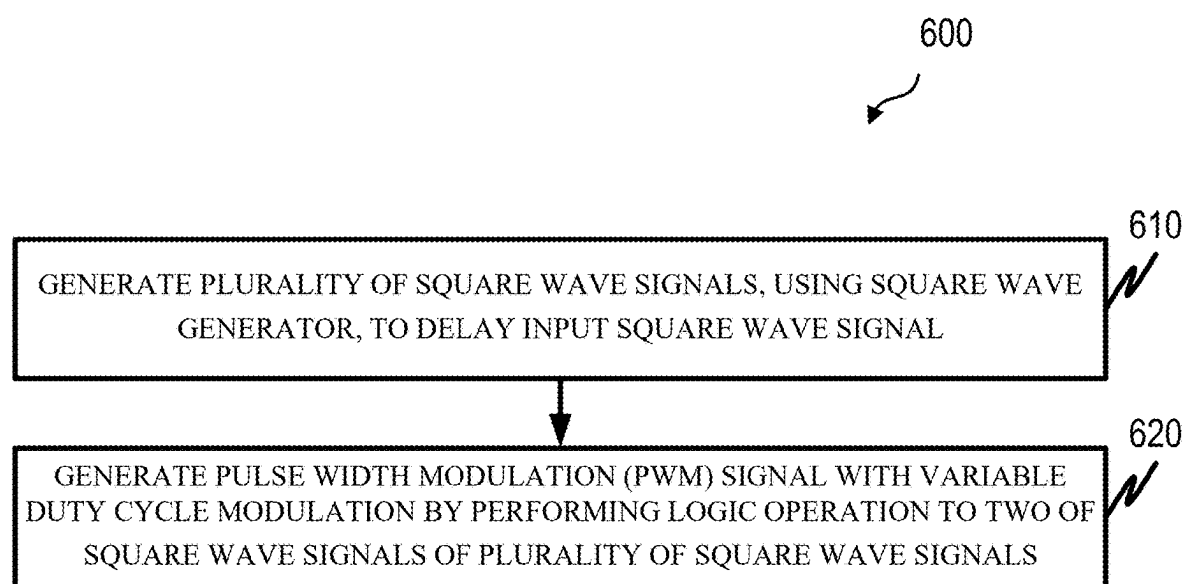


FIG. 6

PULSE WIDTH CONTROL APPARATUS AND METHOD

PRIORITY CLAIM

[0001] This application is a continuation of U.S. patent application Ser. No. 18/366,991, filed on Aug. 8, 2023, which is a continuation of U.S. patent application Ser. No. 17/836,046, filed on Jun. 9, 2022, which claims priority to U.S. Provisional Application No. 63/329,942, filed on Apr. 12, 2022, the contents of each of which are incorporated herein by reference in their entirety.

BACKGROUND

[0002] High-side integrated driver circuits implemented by Gallium Nitride (GaN) High Electron Mobility Transistors (HEMT) enable high-power throughput of various circuitry. HEMTs have a variety of applications including drive operations of discrete power transistors.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures:

[0004] FIG. 1 is a block diagram illustrating an example PWM control circuit in accordance with various embodiments of the present disclosure.

[0005] FIG. 2A is an electrical schematic illustrating an example logic circuit in accordance with various embodiments of the present disclosure.

[0006] FIG. 2B is an electrical schematic illustrating another example logic circuit in accordance with various embodiments of the present disclosure.

[0007] FIG. 2C is an electrical schematic illustrating another example logic 230 circuit in accordance with various embodiments of the present disclosure.

[0008] FIG. 3A is an example PWM control circuit in accordance with various embodiments of the present disclosure.

[0009] FIG. 3B is a series of voltage diagrams illustrating the operation of PWM control circuit over time in accordance with various embodiments of the present disclosure.

[0010] FIG. 4 is an example schematic illustrating an oscillator providing input voltages to the PWM control circuit in accordance with various embodiments of the present disclosure.

[0011] FIG. 5A is another example PWM control circuit in accordance with various embodiments of the present disclosure.

[0012] FIG. 5B is a series of voltage diagrams illustrating the operation of PWM control circuit over time in accordance with various embodiments of the present disclosure.

[0013] FIG. 6 is a process flow diagram illustrating method of generating a PWM signal in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0014] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature

in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0016] Field-effect transistors (FETs) are transistors that operate an electrical device using an electric field. HEMTs, such as Gallium Nitride (GaN) HEMTs, are a type of FET. Due in part to high current density, high breakdown voltage, and low operational resistance, HEMTs are suitable for use in high-power applications. HEMTs, such as GaN HEMTs, have a variety of useful applications including use in pulse width modulation (PWM) control circuits as described herein. PWM circuits are often used to reduce amounts of power consumed by an integrated circuit (IC).

[0017] GaN HEMTs are capable of delivering large amounts of power because of their unique combination of material characteristics including low resistance, high breakdown fields, wide bandgaps (e.g., 3.36 eV for GaN at room temperature), high current density, large conduction band offset, and/or high saturated electron drift velocity. Typically, PWM circuits contain one or more logic circuits having both p-type HEMTs and n-type HEMTs. These PWM circuits, however, can still experience power loss, be large in size, and/or costly due to the number of electrical components therein.

[0018] The circuits and methods described herein utilize only n-type HEMTs to implement logic circuitry, which in turn can generate PWM signals with minimal to no power loss. For example, FIG. 1 is a block diagram illustrating an example PWM control circuit 100 in accordance with various embodiments of the present disclosure. The PWM control circuit 100 can, in some embodiments, be a GaN PWM control circuit made up of GaN integrated circuits such as HEMTs. The PWM control circuit 100 can include any number of n-type logic circuits such as n-type logic circuit 110 and n-type logic circuit X 120. A single voltage source (e.g., power supply) VDD 130 powers PWM control circuit 100. An input signal 140, such as voltage from an oscillator circuit as described in more detail in FIG. 4, is provided to the PWM control circuit 100. The combination of n-type logic circuits (e.g., n-type logic circuit 110 and n-type logic circuit X 120) utilize the input signal 140 to generate an output PWM signal 150 with varying duty cycles. The duty cycle width can be controlled by input signal 140.

[0019] FIGS. 2A-2C are electrical schematics illustrating example logic circuits 210, 220, 230 for integration in PWM

control circuit **100** of FIG. **1** in accordance with various embodiments of the present disclosure. In some embodiments, logic circuits **210**, **220**, **230** are made up of GaN components. For example, FIG. **2A** is an electrical schematic illustrating an example logic **210** circuit in accordance with various embodiments of the present disclosure. Logic circuit **210** is a NOT gate takes its input value and outputs an inverted value of the input value.

[0020] Logic circuit **210** is made up of an enhancement mode HEMT (E-HEMT) **212** and a depletion mode HEMT (D-HEMT) **214** coupled together in series. E-HEMTs and D-HEMTs are example n-type HEMTs. An E-HEMT requires a positive gate voltage to attract electrons towards the gate or an appropriate doping level in a neighboring barrier to attract electrons toward the gate. Electrons attracted toward the gate and enable current flow within the E-HEMT. In other words, an E-HEMT is turned on or activated by pulling the gate terminal thereof to a voltage level higher than a level of a threshold voltage, V_{th} . Conversely, a D-HEMT is in an on state at a zero gate-source voltage. In other words, a D-HEMT is in an on state at the zero gate-source voltage and turns off when a negative gate voltage is applied. As such, it has a negative threshold voltage (e.g., $-1.0V$). Turning back to FIG. **2A**, a source/drain terminal of the D-HEMT **214** is coupled to a voltage source, V_{dd} (e.g., voltage source **130**). Another source/drain terminal of D-HEMT **214** as well as the gate terminal of D-HEMT **214** are coupled to a source/drain terminal of E-HEMT **212**. It is at this node, where an output voltage, v_{out} , is provided by logic circuit **210**. A gate terminal of E-HEMT **212** is coupled to an input voltage, v_{in} . Another source/drain terminal of the E-HEMT **212** is coupled to electrical ground.

[0021] In operation, when the input voltage, v_{in} , is a positive voltage level, the E-HEMT **212** is turned on or activated and operates as a resistor. At this time, the D-HEMT **214** is turned off or deactivated. In a subsequent operation, when the input voltage, v_{in} , is approximately zero, the E-HEMT **212** is turned off or inactive. At this time, the D-HEMT **214** is turned on or activated and operates as a resistor.

[0022] FIG. **2B** is an electrical schematic illustrating another example logic **220** circuit in accordance with various embodiments of the present disclosure. Logic circuit **220** is a NAND gate takes its input value and outputs a logic low (e.g., '0') only if all its inputs are logic highs (e.g., '1'). Otherwise, the NAND gate outputs a logic high (e.g., '1'). Logic circuit **220** is made up of two E-HEMTs **222**, **224** and a D-HEMT **226** coupled together in series. A source/drain terminal of the D-HEMT **226** is coupled to a voltage source, V_{dd} (e.g., voltage source **130**). Another source/drain terminal of D-HEMT **226** as well as the gate terminal of D-HEMT **226** are coupled to a source/drain terminal of E-HEMT **224**. It is at this node, where an output voltage, v_{out} , is provided by logic circuit **220**. A gate terminal of E-HEMT **224** is coupled to a first input voltage, v_a . Another source/drain terminal of the E-HEMT **224** is coupled to a source/drain terminal of E-HEMT **222**. A gate terminal of E-HEMT **222** is coupled to a second voltage source, v_b . Another source/drain terminal of E-HEMT **222** is coupled to ground.

[0023] In operation, when the first input voltage, v_a , is a positive voltage level, the E-HEMT **224** is turned on/activated and operates as a resistor. Similarly, when the second

input voltage, v_b , is a positive voltage level, the E-HEMT **222** is turned on or activated and acts as a resistor. When either E-HEMT **222** or E-HEMT **224** are turned on or activated, the D-HEMT **226** is turned off or deactivated. In a subsequent operation, when the first input voltage, v_a , is approximately zero, the E-HEMT **224** is turned off or deactivated. Similarly, when the second input voltage, v_b , is approximately zero, the E-HEMT **222** is turned off or deactivated. When either of the E-HEMT **222** or E-HEMT **224** are turned off or deactivated, the D-HEMT **226** is turned on or activated and operates as a resistor.

[0024] FIG. **2C** is an electrical schematic illustrating another example logic **230** circuit in accordance with various embodiments of the present disclosure. Logic circuit **230** is a NOR gate takes its input value and outputs a logic high (e.g., '1') only if all its inputs are logic lows (e.g., '0'). Otherwise, the NOR gate outputs a logic low (e.g., '0'). Logic circuit **230** is made up of two E-HEMTs **232**, **234** and a D-HEMT **236**. E-HEMTs **232**, **234** are coupled together in parallel. The D-HEMT **236** is coupled to the E-HEMT **232** in series. A source/drain terminal of the D-HEMT **236** is coupled to a voltage source, V_{dd} (e.g., voltage source **130**). Another source/drain terminal of D-HEMT **236** as well as the gate terminal of D-HEMT **236** are coupled to a source/drain terminal of each of E-HEMT **232** and E-HEMT **234**. It is at this node, where an output voltage, v_{out} , is provided by logic circuit **230**. A gate terminal of E-HEMT **232** is coupled to a first input voltage, v_a . Another source/drain terminal of the E-HEMT **232** is coupled to a ground. A gate terminal of E-HEMT **234** is coupled to a second voltage source, v_b . Another source/drain terminal of E-HEMT **234** is coupled to ground.

[0025] In operation, when the first input voltage, v_a , is a positive voltage level, the E-HEMT **232** is turned on/activated and operates as a resistor. Similarly, when the second input voltage, v_b , is a positive voltage level, the E-HEMT **234** is turned on or activated and acts as a resistor. When either E-HEMT **232** or E-HEMT **234** are turned on or activated, the D-HEMT **236** is turned off or deactivated. In a subsequent operation, when the first input voltage, v_a , is approximately zero, the E-HEMT **232** is turned off or deactivated. Similarly, when the second input voltage, v_b , is approximately zero, the E-HEMT **234** is turned off or deactivated. When either of the E-HEMT **232** or E-HEMT **234** are turned off or deactivated, the D-HEMT **236** is turned on or activated and operates as a resistor.

[0026] FIG. **3A** is an example PWM control circuit **300** in accordance with various embodiments of the present disclosure. The PWM control circuit **300**, in some embodiments, is a GaN device having GaN components. For example, PWM control circuit **300** includes a number of logic circuits **310**, **320**, **330**, **340** that are fully integrated in a GaN chip. Logic circuits **310**, **320**, and **340** are NOT gates each having substantially equivalent components, couplings, and operations as previously described in FIG. **2A**. In other words, logic circuit **310** includes an E-HEMT **312** and a D-HEMT **314**, where D-HEMT **314** is substantially equivalent to D-HEMT **214** and E-HEMT **312** is substantially equivalent to E-HEMT **212**. Similarly, logic circuit **320** includes an E-HEMT **322** and a D-HEMT **324**, where D-HEMT **324** is substantially equivalent to D-HEMT **214** and E-HEMT **322** is substantially equivalent to E-HEMT **212**. Likewise, logic circuit **340** includes an E-HEMT **342** and a D-HEMT **344**,

where D-HEMT **344** is substantially equivalent to D-HEMT **214** and E-HEMT **342** is substantially equivalent to E-HEMT **212**.

[0027] Logic circuit **330** is a NAND gate having substantially equivalent components, couplings, and operations as previously described in FIG. 2B. In other words, logic circuit **330** includes E-HEMTs **332**, **334** and a D-HEMT **336**, where E-HEMT **332** is substantially equivalent to E-HEMT **222**, E-HEMT **334** is substantially equivalent to E-HEMT **224**, and D-HEMT **336** is substantially equivalent to D-HEMT **226**.

[0028] Logic circuits **310**, **320** provide input voltages **velk1** and **velkb1**, respectively, to logic circuit **330**. In turn, logic circuit **330** provides an input voltage **vnand** to logic circuit **340**. Logic circuit **340** outputs a PWM signal **350** having a duty cycle based on the duty cycles of voltages input to logic circuit **310**, **320**. For example, logic circuit **310** is coupled to logic circuit **330** and provides an output voltage (e.g., **velk1**) via a node coupling E-HEMT **312** and D-HEMT **314** to a gate terminal of E-HEMT **334** of logic circuit **330** (e.g., input voltage **va** of logic circuit **220**). Logic circuit **320** is also coupled to logic circuit **330** and provides an output voltage (e.g., **velk2**) via a node coupling E-HEMT **322** and D-HEMT **324** to a gate terminal of E-HEMT **332** of logic circuit **330** (e.g., input voltage **vb** of logic circuit **220**). Logic circuits **310**, **320** each receive an input voltage (e.g., input voltage **vin** of circuit **210**) **vn18**, **vn25**, respectively, from an external voltage source. Logic circuit **330** outputs an output voltage (e.g., output voltage **vout** of logic circuit **220**) to a gate terminal of E-HEMT **342** of logic circuit **340**. Logic circuit **340** in turn outputs a PWM signal **150** that is based on the duty cycles of input voltages **vn18**, **vn25** of logic circuits **310**, **320**, respectively.

[0029] FIG. 3B is a series of voltage diagrams **360**, **370**, **380**, **390** illustrating the operation of PWM control circuit **300** over time in accordance with various embodiments of the present disclosure. The voltages illustrated in voltage diagrams **360**, **370**, **380**, **390** are for illustrative purposes and demonstrate voltage as a function of time with logic amplitudes of a logic high (e.g., '1') or a logic low (e.g., '0'). Voltage diagram **360** illustrates an input voltage **vn18** to logic circuit **310** having a duty cycle (e.g., ratio of the pulse width to the period) of approximately 40%-50%. Voltage diagram **370** illustrates an input voltage **vn25** to logic circuit **320** having a duty cycle of approximately 40%-50%. As previously discussed, input voltages **vn18** and **vn25** are provided by one or more external voltage sources. The two input voltages **vn18** and **vn25** are combined together using a NAND logic operation via logic circuit **330**. Voltage diagram **380** illustrates the output voltage **vnand** of logic circuit **330**. As previously discussed, a NAND gate takes its input value and outputs a logic low (e.g., '0') only if all its inputs are logic highs (e.g., '1'). Otherwise, the NAND gate outputs a logic high (e.g., '1'). This operation is illustrated in voltage diagram **380**. Voltage diagram **390** illustrates the PWM signal **350** output by logic gate **330**. The voltage diagram **390** is an inverted version of voltage diagram **380**, as logic gate **330** performs the logic operation of NOT, which inverts the input voltage. The duty cycle associated with voltage diagram **390** is approximately 10%-15%.

[0030] FIG. 4 is an example schematic illustrating an oscillator **400** providing input voltages **vn18**, **vn25** to the PWM control circuit **300** in accordance with various embodiments of the present disclosure. Input voltages **vn18**,

vn25 can be provided by an external power source, such as an oscillator. Oscillator **400** includes a number of logic circuits **410**, **420**, **430**, **440**, **450** as well as additional logic circuits not illustrated in FIG. 4 that are used to generate voltages **vn18**, **vn25**. Logic circuit **410** is a NAND gate having substantially equivalent components, couplings, and operations as previously described in FIG. 2B. In other words, logic circuit **410** includes E-HEMTs **412**, **414** and a D-HEMT **416**, where E-HEMT **412** is substantially equivalent to E-HEMT **222**, E-HEMT **414** is substantially equivalent to E-HEMT **224**, and D-HEMT **416** is substantially equivalent to D-HEMT **226**. Logic circuits **420**, **430**, **440**, **450** are each NOT gates each having substantially equivalent components, couplings, and operations as previously described in FIG. 2A. In other words, logic circuit **420** includes an E-HEMT **422** and a D-HEMT **424**, where D-HEMT **424** is substantially equivalent to D-HEMT **214** and E-HEMT **422** is substantially equivalent to E-HEMT **212**. Similarly, logic circuit **430** includes an E-HEMT **432** and a D-HEMT **434**, where D-HEMT **434** is substantially equivalent to D-HEMT **214** and E-HEMT **432** is substantially equivalent to E-HEMT **212**. Likewise, logic circuit **440** includes an E-HEMT **442** and a D-HEMT **444**, where D-HEMT **444** is substantially equivalent to D-HEMT **214** and E-HEMT **442** is substantially equivalent to E-HEMT **212**. Correspondingly, logic circuit **450** includes an E-HEMT **452** and a D-HEMT **454**, where D-HEMT **454** is substantially equivalent to D-HEMT **214** and E-HEMT **452** is substantially equivalent to E-HEMT **212**.

[0031] Logic circuit **410** receives a first input voltage, **Ven**, (e.g., **va** of logic circuit **220**), at a gate terminal of E-HEMT **414**. Logic circuit **420** also received a second input voltage, **vn31**, (e.g., **vb** of logic circuit **220**) output from logic circuit **450** at a gate terminal of E-HEMT **412**. Logic circuit **410** is coupled to logic circuit **420** and provides its output voltage **vn1** (e.g., **vout** of logic circuit **220**) to a gate terminal of E-HEMT **422** (e.g., **vin** of logic circuit **210**). Logic circuit **420** is coupled to logic circuit **430** and provides its output voltage **vn2** (e.g., **vout** of logic circuit **210**) to a gate terminal of E-HEMT **432** (e.g., **vin** of logic circuit **210**). Logic circuit **430** is coupled to logic circuit **440** and provides its output voltage **vn3** (e.g., **vout** of logic circuit **210**) to a gate terminal of E-HEMT **440** (e.g., **vin** of logic circuit **210**). Similarly, logic circuit **440** is coupled to another series of logic circuit not illustrated in FIG. 4 and is represented by series of continuing dots and provides its output voltage **v4** (e.g., **vout** of logic circuit **210**) to the next coupled logic circuit. Such a series of logic circuits continues until logic circuit **450**. Logic circuit **450** is coupled to a logic circuit not illustrated in FIG. 4 and receives an input voltage **vn30** at a gate terminal of E-HEMT **452**. Logic circuit **450** is coupled to logic circuit **410**, as previously discussed, and provides a voltage **vn30** (e.g., **vout** of logic circuit **210**). Based on the desired input waveform characteristics for PWM control circuit **300**, voltages from the oscillator **400** can be provided at various connection points on the series of logic circuits within oscillator **400**. For example, voltages **vn18** and **vn25** can be taken from an eighteenth and twenty-fifth logic circuit, respectively, within oscillator **400**. Such voltages can be provided to PWM control circuit **300** as described in detail in FIGS. 3A-3B.

[0032] It can be appreciated by those of ordinary skill in the art, however, that oscillator **400** is merely one example circuit that can provide voltage and any voltage source can

be utilized to generate and provide voltage to PWM control circuit 300. Additionally, the component makeup of an oscillator can differ from that illustrated in FIG. 4.

[0033] FIG. 5A is another example PWM control circuit 500 in accordance with various embodiments of the present disclosure. PWM control circuit 500 can have similar components as PWM control circuit 300, with an additional logic circuit 510 coupled thereto. The PWM control circuit 500, in some embodiments, is a GaN device having GaN components. Logic circuit 510 is a NOT gate having substantially equivalent components, couplings, and operations as previously described in FIG. 2A. In other words, logic circuit 510 includes an E-HEMT 512 and a D-HEMT 514, where D-HEMT 514 is substantially equivalent to D-HEMT 214 and E-HEMT 512 is substantially equivalent to E-HEMT 212. Logic circuit 510 receives an input voltage vnot (e.g., vin of logic circuit 210) from PWM control circuit 300 and outputs a PWM signal 520 at a source/drain terminal coupling point of E-HEMT 512 and D-HEMT 514. In the embodiment illustrated in FIG. 5A, PWM control circuit 500 receives different input voltages vn14, vn29 from an external voltage source, rather than vn18, vn25 as described in FIG. 3A. These external voltages can be supplied, in some embodiments, from serial logic circuits of oscillator 400 of FIG. 4.

[0034] FIG. 5B is a series of voltage diagrams 530, 540, 550, 560, 570 illustrating the operation of PWM control circuit 500 over time in accordance with various embodiments of the present disclosure. The voltages illustrated in voltage diagrams 530, 540, 550, 560, 570 are for illustrative purposes and demonstrate voltage as a function of time with logic amplitudes of a logic high (e.g., '1') or a logic low (e.g., '0'). Voltage diagram 530 illustrates an input voltage vn14 to logic circuit 310 having a duty cycle (e.g., ratio of the pulse width to the period) of approximately 40%-50%. Voltage diagram 540 illustrates an input voltage vn29 to logic circuit 320 having a duty cycle of approximately 40%-50%. As previously discussed, input voltages vn14 and vn29 are provided by one or more external voltage sources. The two input voltages vn14 and vn29 are combined together using a NAND logic operation via logic circuit 330. Voltage diagram 550 illustrates the output voltage vnot of logic circuit 330. As previously discussed, a NAND gate takes its input value and outputs a logic low (e.g., '0') only if all its inputs are logic highs (e.g., '1'). Otherwise, the NAND gate outputs a logic high (e.g., '1'). This operation is illustrated in voltage diagram 550. Voltage diagram 560 illustrates an output voltage vnot of logic circuit 340. The voltage diagram 560 is an inverted version of voltage diagram 550, as logic gate 330 performs the logic operation of NOT, which inverts the input voltage. The duty cycle associated with voltage diagram 560 is approximately 15%-20%. Voltage diagram 570 illustrates the PWM signal 550 output by logic gate 510. The voltage diagram 570 is an inverted version of voltage diagram 560, as logic gate 510 performs the logic operation of NOT, which inverts the input voltage. The duty cycle associated with voltage diagram 560 is approximately 70%-75%.

[0035] FIG. 6 is a process flow diagram 600 illustrating method of generating a PWM signal in accordance with various embodiments of the present disclosure. While FIG. 6 is described here with reference to previously described structures for ease in understanding, it is understood that the method applies to many other structures as well. A plurality

of square wave signals (e.g., vn1, vn2, vn3, vn4, . . . vn14, . . . , vn18, . . . , vn25, . . . , vn29, vn30, vn31 of oscillator 400) are generated at step 610, using a square wave generator such as oscillator 400, to delay an input square wave signal (e.g., Ven). A PWM signal such as PWM signals 350, 520 are generated at step 620 with variable duty cycle modulation (e.g., voltage diagram 390 v. voltage diagram 570) by performing a logic operation (e.g., NOT, NOR, or NAND) to two of square wave signals (e.g., vn14, vn18 and/or vn25, vn29) of the plurality of square wave signals (e.g., vn1, vn2, vn3, vn4, . . . vn14, . . . , vn18, . . . , vn25, . . . , vn29, vn30, vn31 of oscillator 400).

[0036] Use of the various circuits and methods as described herein can provide a number of advantages. For example, the various circuits and methods described herein does not require a sawtooth generator, level shifter, or comparator to provide an input voltage to the PWM control circuits. Additionally, the circuits and methods provide for a fully integrated design that reduces switching loss from assembled complementary metal-oxide-semiconductor (CMOS) control integrated circuit (IC) and GaN power device. Additionally, the circuits and methods described herein do not require additional passive components, thereby improving performance and minimizing the size of the device.

[0037] In one embodiment, a system includes a square wave generator and a logic device. The square wave generator is configured to delay a input square wave signal to generate a plurality of square wave signals. The logic device is configured to perform a logic operation to two of square wave signals of the plurality of square wave signals that in turn generates a PWM signal having a duty cycle corresponding to the two square wave signals.

[0038] In another embodiment, a device includes a plurality of logic circuits comprising a plurality of E-HEMTs and a plurality of D-HEMTs. The plurality of logic circuits are configured to generate a PWM signal based on voltage inputs to the plurality of logic circuits.

[0039] In yet another embodiment, a plurality of square wave signals are generated, using a square wave generator, to delay an input square wave signal. A PWM signal is generated with variable duty cycle modulation by performing a logic operation to two of square wave signals of the plurality of square wave signals.

[0040] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A system comprising:

a logic device configured to perform a logic operation to a signal generated by a oscillator circuit, wherein the logic operation generates a pulse width modulation (PWM) signal corresponding to the signal.

2. The system of claim 1, wherein the logic device comprises a plurality of enhancement mode high electron mobility transistors (E-HEMTs) and a plurality of depletion mode high electron mobility transistors (D-HEMTs).

3. The system of claim 1, wherein the logic device is configured to perform the logic operation to two signals generated by the oscillator circuit, and the logic device comprises:

a first logic circuit and a second logic circuit configured to receive the two signals; and

a third logic circuit configured to output the PWM signal.

4. The system of claim 3, wherein:

the first logic circuit is configured to receive a first signal of the two signals, the first logic circuit comprising a first E-HEMT coupled in series to a first D-HEMT;

the second logic circuit is configured to receive a second signal of the two signals, the second logic circuit comprising a second E-HEMT coupled in series to a second D-HEMT;

the logic device further comprises a fourth logic circuit coupled to the first logic circuit and the second logic circuit, the fourth logic circuit comprising a third E-HEMT and a fourth E-HEMT coupled together in series with a third D-HEMT; and

the third logic circuit is coupled to the fourth logic circuit, the third logic circuit comprising a fifth E-HEMT and a fourth D-HEMT.

5. The system of claim 4, wherein the logic device further comprises a fifth logic circuit coupled between the third logic circuit and the fourth logic circuit, the fifth logic circuit comprising a sixth E-HEMT and a fifth D-HEMT.

6. The system of claim 2, wherein the plurality of E-HEMTs and the plurality of D-HEMTs are n-type transistors.

7. The system of claim 1, wherein the logic operation comprises at least one of a NOT gate, a NAND gate, or a NOR gate.

8. The system of claim 1, further comprising a signal generator configured to delay an input signal to generate the signals, and the signal generator comprises the oscillator circuit.

9. The system of claim 1, wherein the logic device comprises Gallium Nitride (GaN) components.

10. A device comprising:

a plurality of logic circuits configured to generate an output signal based on signals generated by an oscillator circuit and to the plurality of logic circuits, wherein the plurality of logic circuits comprise:

first and second logic circuits configured to receive the signals.

11. The device of claim 10, wherein the plurality of logic circuit comprises a plurality of high electron mobility transistors (HEMTs), the output signal is a pulse width modulation (PWM) signal, and the plurality of logic circuits further comprises a third logic circuit configured to output the PWM signal.

12. The device of claim 11, wherein the signals to the plurality of logic circuits comprise at least two signals generated by a signal generator and the PWM signal has a duty cycle signal that is based on duty cycles of the at least two signals.

13. The device of claim 11, wherein:

the first logic circuit is configured to receive a first signal of the signals, the first logic circuit comprising a first

E-HEMT (enhancement-HEMT) of the plurality of E-HEMTs coupled in series to a first D-HEMT (depletion-HEMT) of the plurality of D-HEMTs;

the second logic circuit is configured to receive a second signal of the signals, the second logic circuit comprising a second E-HEMT of the plurality of E-HEMTs coupled in series to a second D-HEMT of the plurality of D-HEMTs;

the plurality of logic circuits further comprise a fourth logic circuit coupled to the first logic circuit and the second logic circuit, the fourth logic circuit comprising a third E-HEMT and a fourth E-HEMT of the plurality of E-HEMTs coupled together in series with a third D-HEMT of the plurality of D-HEMTs; and

the third logic circuit is coupled to the fourth logic circuit, the third logic circuit comprising a fifth E-HEMT of the plurality of E-HEMTs and a fourth D-HEMT of the plurality of D-HEMTs.

14. The device of claim 13, wherein the plurality of logic circuits further comprises a fifth logic circuit coupled between the third logic circuit and the fourth logic circuit, the fifth logic circuit comprising a sixth E-HEMT of the plurality of E-HEMTs and a fifth D-HEMT of the plurality of D-HEMTs.

15. The device of claim 11, wherein the plurality of E-HEMTs and the plurality of D-HEMTs are n-type transistors.

16. The device of claim 10, wherein the signal generator comprises the oscillator circuit.

17. A method, comprising:

generating an output signal by performing a logic operation to a signal generated by an oscillator circuit, wherein the output signal is generated by:

receiving, by a first logic circuit, the signal; and

generating, by a third logic circuit, the output signal.

18. The method of claim 17, wherein the output signal is a pulse width modulation (PWM) signal with variable duty cycle modulation, the PWM signal is generated by:

receiving, by the first logic circuit and a second logic circuit, two signals generated by the oscillator circuit;

generating, by the first logic circuit and the second logic circuit, clocking voltages configured to drive a fourth logic circuit;

performing, by the fourth logic circuit, a first logic operation to generate a first voltage output; and

generating, by the third logic circuit, the PWM signal by performing a second logic operation on the first voltage output.

19. The method of claim 18, wherein:

the first logic circuit comprises a first E-HEMT coupled in series to a first D-HEMT;

the second logic circuit comprises a second E-HEMT coupled in series to a second D-HEMT;

the fourth logic circuit is coupled to the first logic circuit and the second logic circuit and comprises a third E-HEMT and a fourth E-HEMT coupled together in series with a third D-HEMT; and

the third logic circuit is coupled to the fourth logic circuit and comprises a fifth E-HEMT and a fourth D-HEMT.

20. The method of claim 18, wherein the PWM signal is further generated by performing, by a fifth logic circuit coupled between the third logic circuit and the fourth logic circuit, a third logic operation.

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